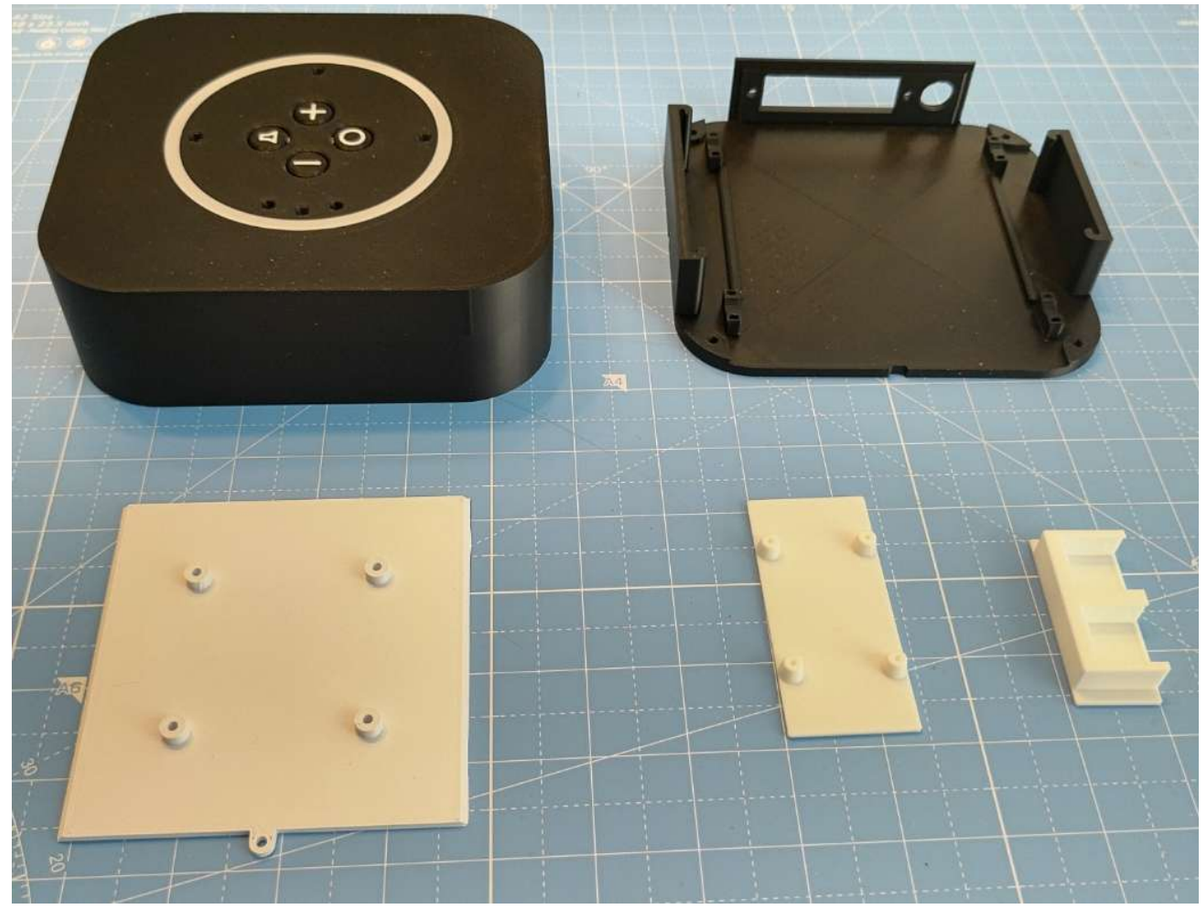
FPH Sat-1
Amp assembly



Printed Parts: (Small case shown)

1. **Top (Large)**
2. **Amp Base (Large)**
3. **Amp PCB Mount (Large)**
4. **Wago Connector Insert**
5. **Buck Converter Insert**

Filament

Standard settings
No supports are required
The LED ring has special print settings
Use Transparent or Basic White

Power

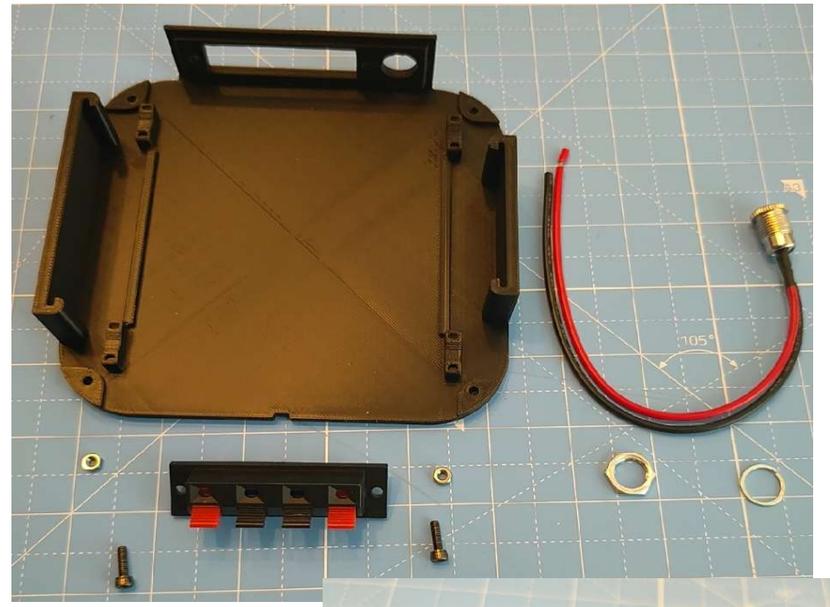
Testing was done using a DC 12V 3 Amp
& 24V 4 Amp power adapters.
Check specifications of PCB if using other PSU
USE AT YOUR OWN RISK!

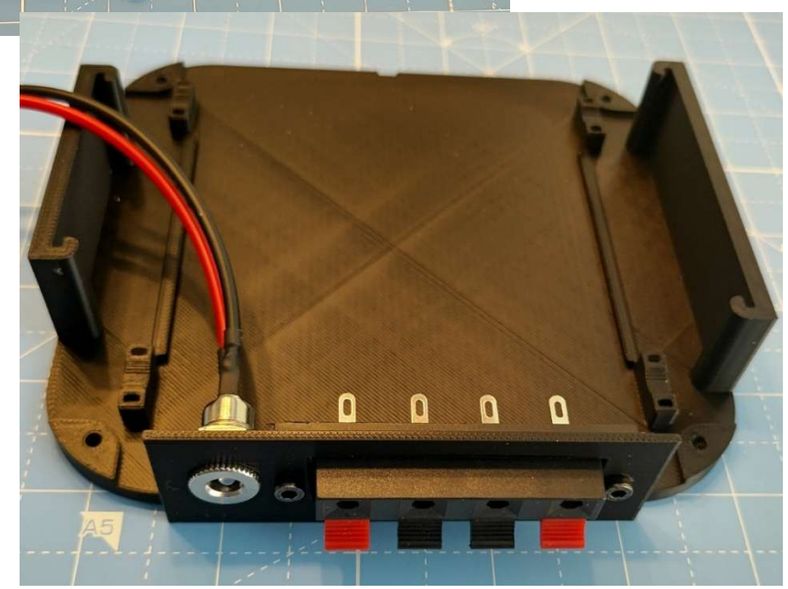
FPH Sat-1
Amp assembly

BOM (links to sources at end of guide)

1. **M3x4 screws - 6**
2. **M3x4 to M3x8 screws – 2**
   a. **Sat1**
3. **M3x8 to M3x10 screws – 8**
   a. **Sat1 & Speaker Term Block**
4. **M3 nuts – 2 (Speaker Term Block)**
5. **M2x4 screws – 4 (Buck converter)**
6. **16 or 18 AWG wire (red / black)**
7. **Ferrules crimp connectors**
   a. **(optional)**
8. **Wago 221-413 – 2**
9. **Buck Converter**
10. **Amp Small**
    a. **TPA3110 (2 versions) & TPA3116D2**
11. **Amp Large**
    a. **Zk-2202-W Class D Amp (TPA3251)**
12. **DC power jack**
13. **Speaker Terminal Block**
14. **2.8 female spade connector**
15. **Right Angle USB-C Cable (options)**
16. **3.5mm Audio Cable (options)**

FPH Sat-1
Amp assembly





Assembly

Assembly is the same for both sizes

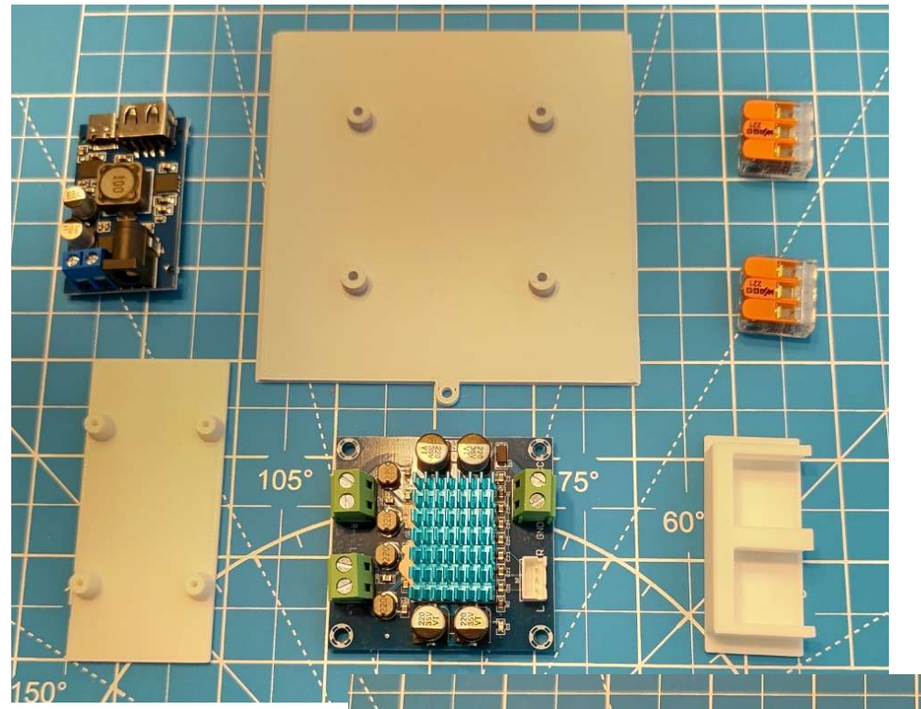
DC Power Jack

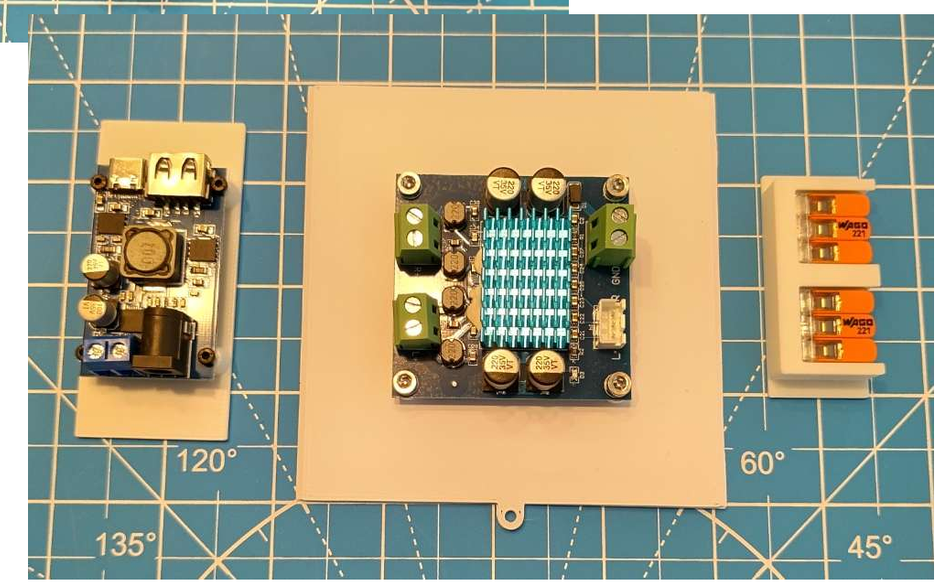
1. **Insert into base**
2. **Add washer and nut**
3. **Tighten with fingers**
4. **Use wrench to finish tightening**

Speaker Terminal Block

1. **Insert Block with spring clip down**
    a. **Insert from inside the Amp Base**
2. **Insert M3 screws from rear and add nut**
3. **Lightly Tighten**
4. **Add second M3 screw and add nut**
5. **Tighten both securely**

FPH Sat-1
Amp assembly





Wago Connectors

1. Insert the two 221-413 into slots
2. Slide the mount into the side

Buck Converter

1. Mount converter using the 4 M2x4 screws
2. Slide the converter into the with USB-C port facing the rear

Amp

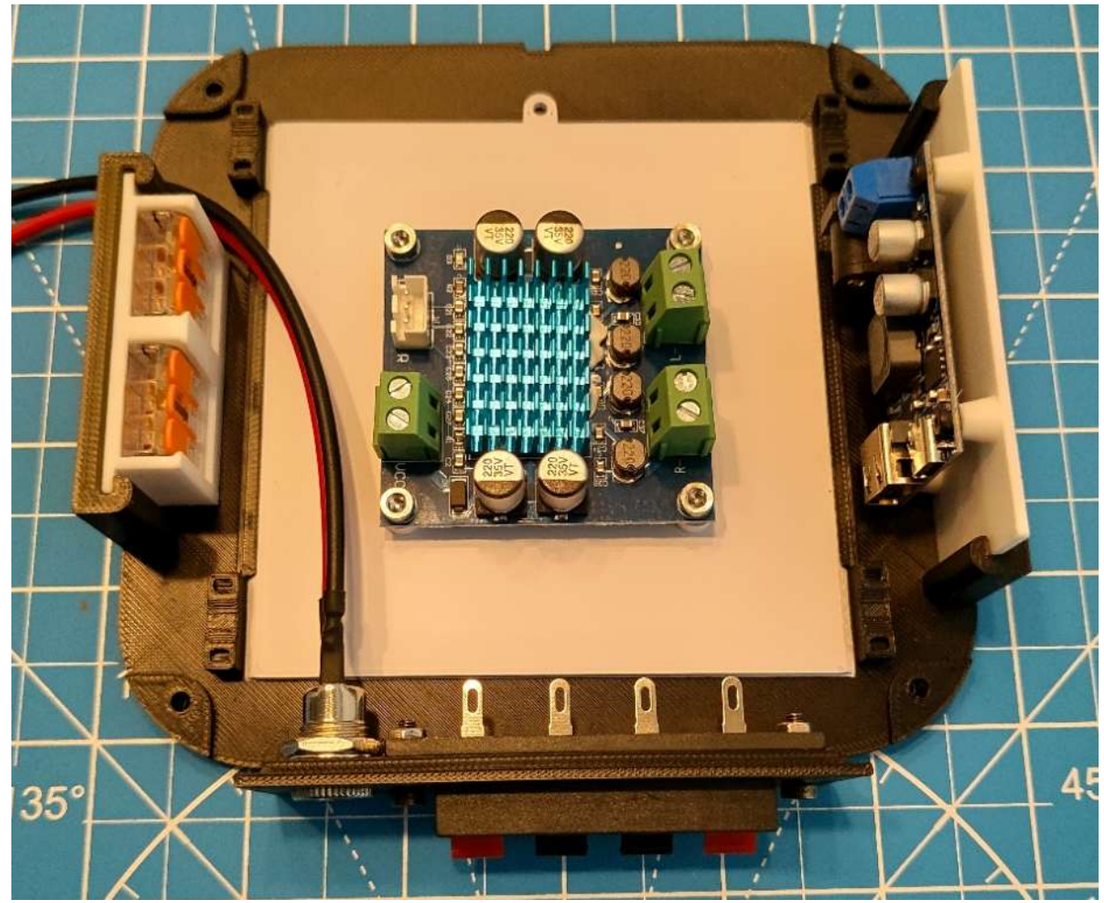
1. Mount the Amp on the AMP PCB mount
2. Use 4 M3x4 screws
3. Slide the PCB mount on to base

FPH Sat-1
Amp assembly

Wire Lengths (longest)

Small Case
1.Buck Converter to Wago – 200mm
2.Amp to Wago – 50mm
3.Amp to Speaker Block – 100mm

Large Case
1.Buck Converter to Wago – 250mm
2.Amp to Wago – 50mm
3.Amp to Speaker Block – 170mm



FPH Sat-1
Amp assembly

Cut Power wires to fit
Cut wires to length as needed

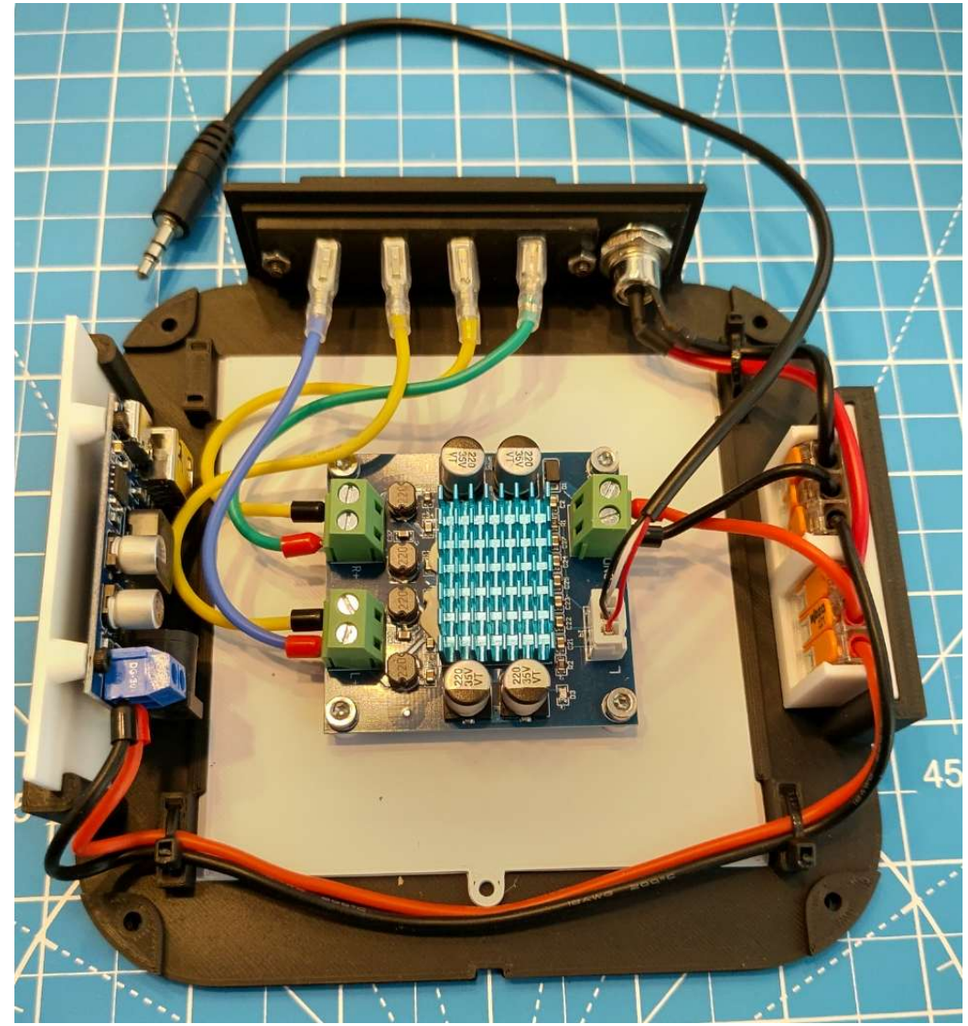
Crimp 2.8 female spade connectors

If using ferrules connectors, crimp now
Note: Makes inserting wire into connectors easier.

Caution:

<u>Be sure to double check when connecting wires.</u>

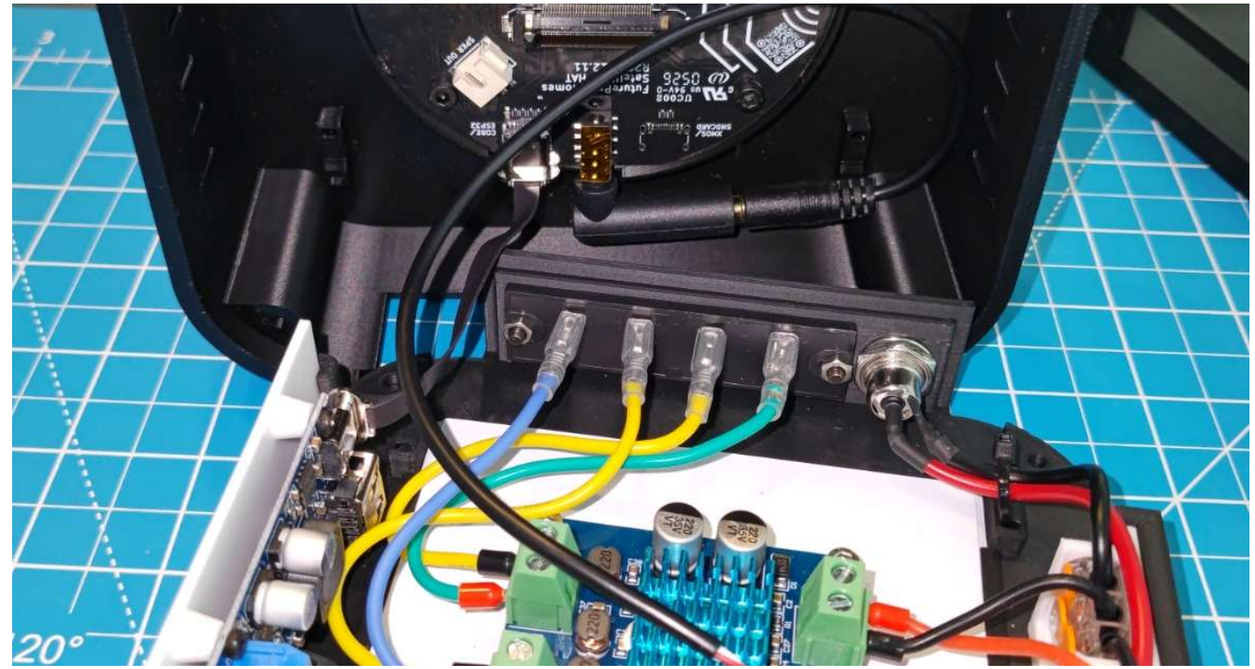
There are zip ties blocks at the corners to tie down wires if needed.



FPH Sat-1
Amp assembly

Once all the wires are connected.
Recheck each connection.

Attach the USC-C cable to Sat1.
Attach 3.5mm audio cable.
Attach speakers to speaker block.

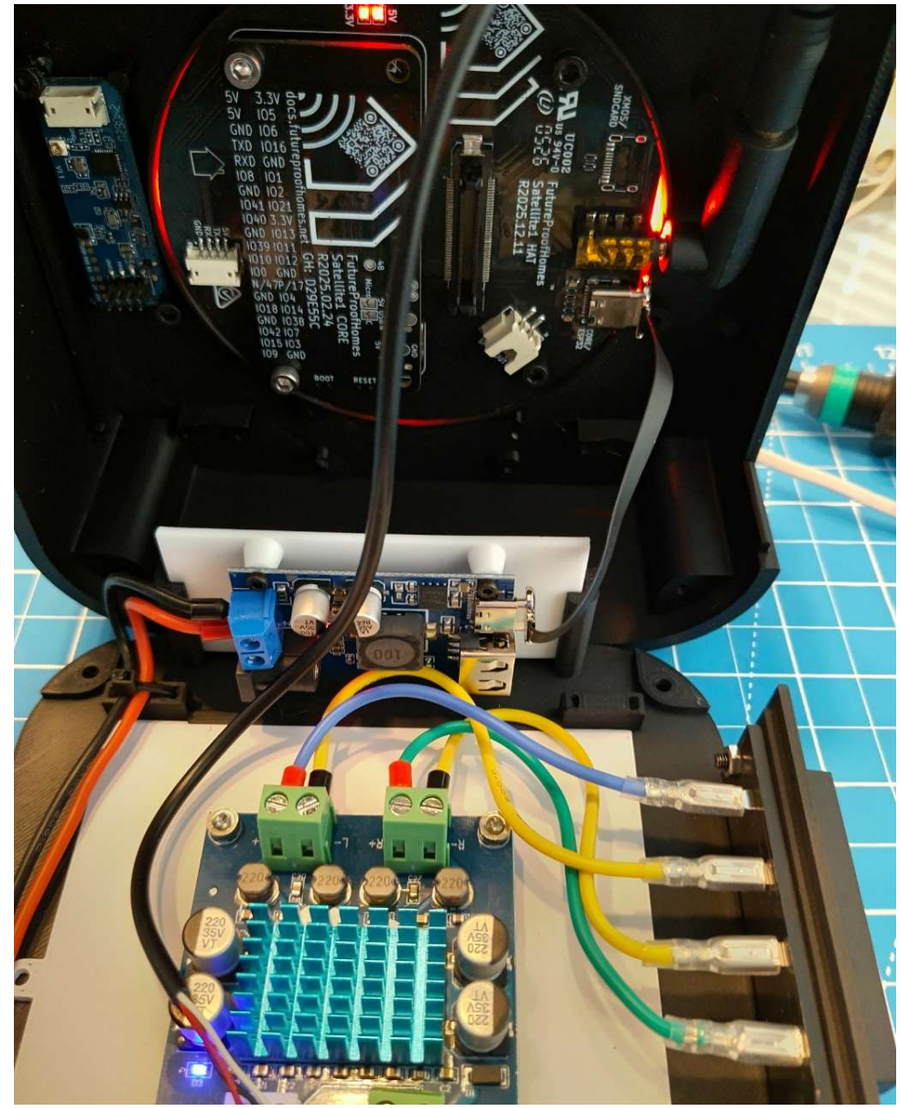


FPH Sat-1
Amp assembly

Amp Large – turn knob to off position
If all looks good, plug in power.
Amp Large – turn on amp, slowly turn up

There should be lights on PCBs and Sat1
There may be an initial pop on the speakers
If there are no lights on PCBs, check Wago's
If there is no audio, check wiring at block

Finally
Slide to the two parts together
Use 4 M3x8 screws to secure bottom to top,



FPH Sat-1
Amp assembly

New LD2450 and Wi-Fi antenna mounts

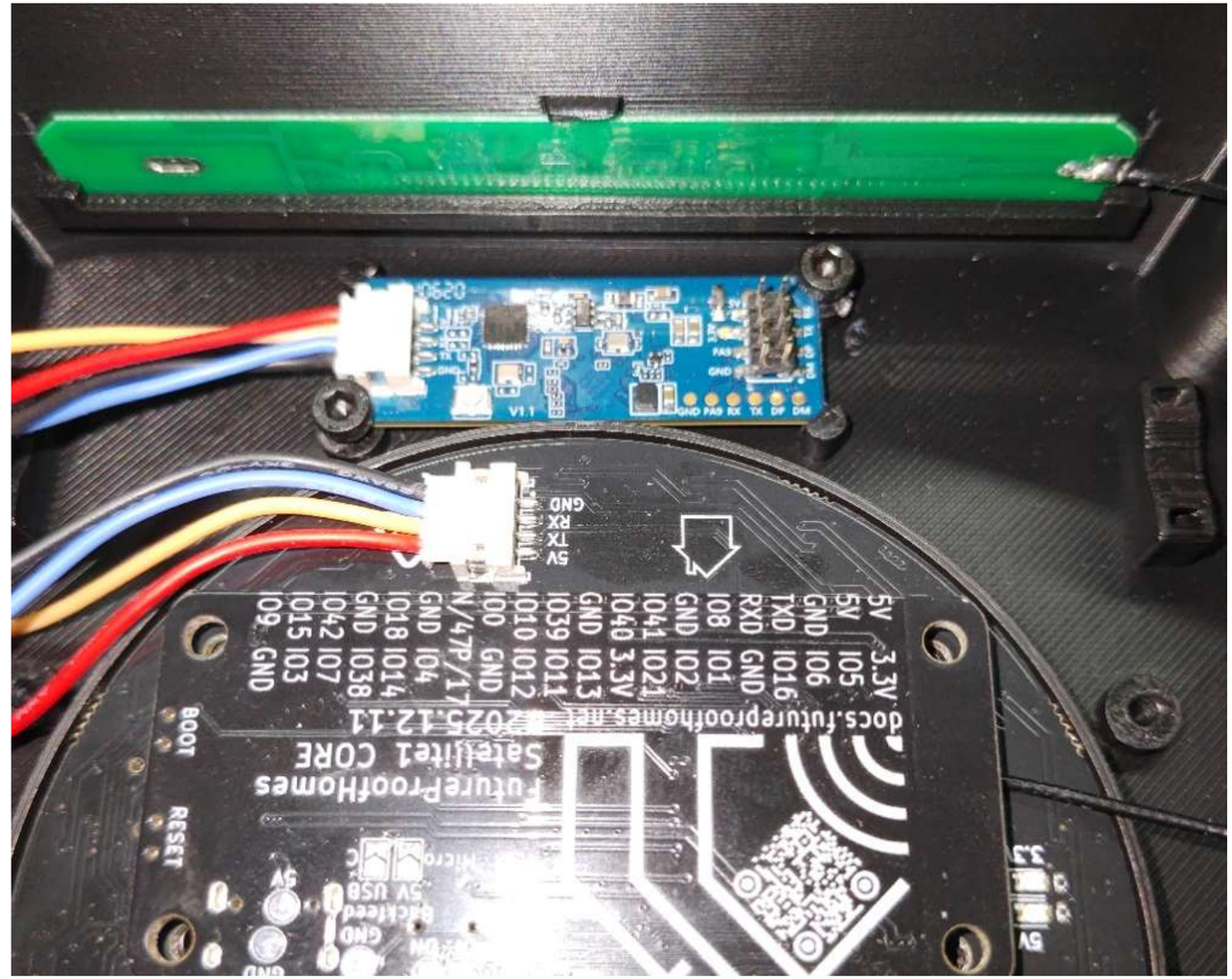
There are 4 mounting places for the LD2450
The board is held in be 2 M3x4 screws

1. **Get the screws started in the corners**
2. **Slide the board in**
3. **Set it in cutouts in opposite corners**
4. **Tighten screws until contact**

Note: if wiring a cable the RX goes to TX
and TX goes to RX

he Wi-Fi antenna slides in the slot and
the clip at the top holds it in place.

To remove the antenna gently pull on the
middle and pull up on the end



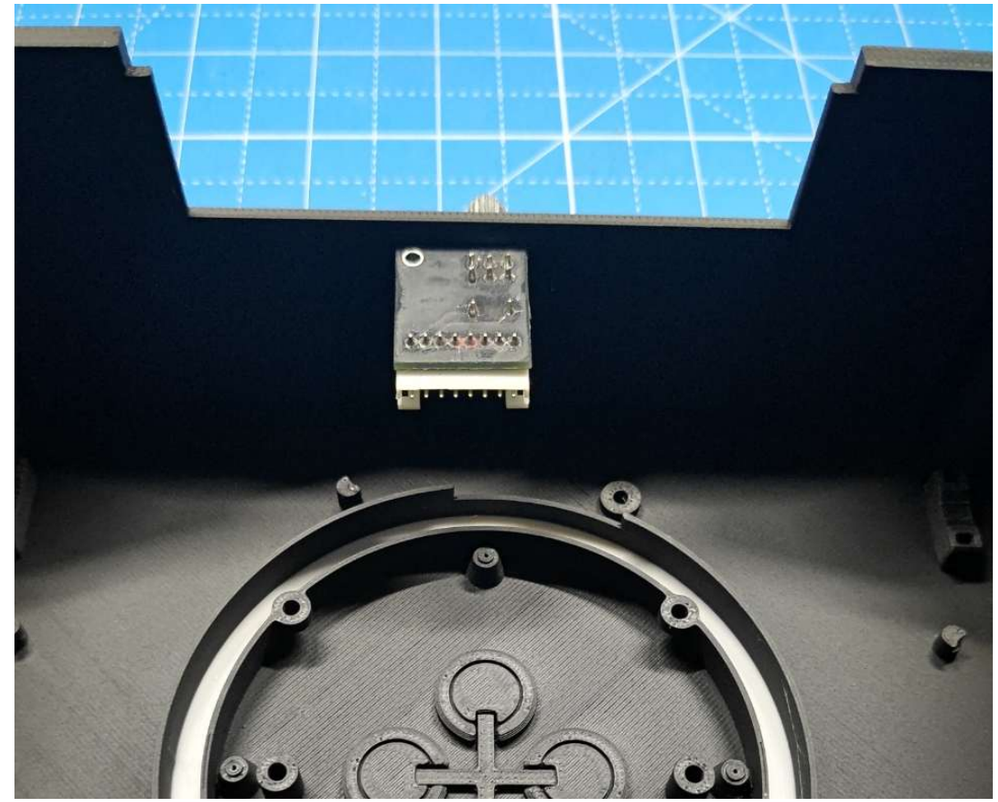
FPH Sat-1
Amp assembly

TPA3251 Volume Knob Mount

The mount has moved to the rear for a cleaner look.

There is a knotch at the top for the tab on the knob.

1. **Mount the knob with the PCB on the top**
2. **Press the knob into the hole**
    a. **It should snap in place**
3. **Tighten with wrench**



FPH Sat-1
Amp assembly

Source Links (what I used)

M3 Screws and Nuts and M2 Screws
Any source will work

Wire – 18 AWG (multi color, silicone)
https://www.amazon.com/dp/B09Y856K8J?ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_21&th=1

Ferrules Connectors (Optional)
https://www.amazon.com/dp/B0FKYK38RZ?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1
Crimper
https://www.amazon.com/dp/B0D7ZVC29K?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

FPH Sat-1
Amp assembly

Wago (Recommend buying a multi pack)
Multi-pack
https://www.amazon.com/dp/B0BN6MBR7X?ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_5

221-413 (alone)
https://www.amazon.com/WAGO-221-413-Lever-Nuts-3-Conductor-Connectors/dp/B07W94XJSN/ref=sr_1_1?crid=2O4I4YXLM5DKC&dib=eyJ2IjoiMSJ9.8G0diYVfWMiQJVcCrLRBeJPmnQuniNHTXBQg8uYtRSu9JXM2ifpG0bCjhoBeuX7ptKaH3WaMrSANeaO8dis6w9rc-e2NZMHdxRsW-CsT-6toGhLNJHUpFwSpulw4KGRKmuu54BbPJitgI0n5IVeJd9LJNi9FKSNk2V9D7yvYF5tRuECZmMoM7tD2URH5Dy4Hc-CzaiEcEK7cttnuptYC2YM9O3DOR2jNL6Kvow7s8mTYaRdC46nEnbE1_OZSU8yle3wI6YhkFuRFnfPCyyoPP7WagPLFG0aDMnOWwuSVJhc.bfXd7ykWwURaMsIUhVobx901-VUtyxUBJU5hjRk-ZdE&dib_tag=se&keywords=wago%2B221-413&qid=1772046604&s=hi&sprefix=wago%2B221-41%2Ctools%2C256&sr=1-1&th=1

FPH Sat-1
Amp assembly

Buck Converter
https://www.amazon.com/dp/B0D146WV15?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

Amp – Small without connectors (TPA3110)
https://www.amazon.com/dp/B0CY9C1936?ref=ppx_yo2ov_dt_b_fed_asin_title

Amp – Small with connectors (TPA3110)
https://www.aliexpress.us/item/3256807803112323.html?spm=a2g0o.order_list.order_list_main.5.51cf1802Utnnov&gatewayAdapt=glo2usa

AMP – Small with 3.5mm audio jack (TPA3116D2) - RECOMMENDED
https://www.amazon.com/dp/B08NFCCGC2?ref=ppx_yo2ov_dt_b_fed_asin_title

Amp – Large – (TPA3251) - RECOMMENDED
https://www.amazon.com/dp/B0DSPLN2QC?ref=ppx_yo2ov_dt_b_fed_asin_title

FPH Sat-1
Amp assembly

DC Power Jack
https://www.amazon.com/dp/B0DLKN8J7M?ref=ppx_yo2ov_dt_b_fed_asin_title

Speaker Terminal Block
https://www.amazon.com/dp/B08YMZ88Z5?ref=ppx_yo2ov_dt_b_fed_asin_title

Spade Connector
https://www.amazon.com/dp/B07FCFH84Q?ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_28&th=1

USB-C Cables (These were used and worked)
https://www.amazon.com/dp/B0G439K5W9?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

https://www.amazon.com/dp/B0FJF1M4CT?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

https://www.amazon.com/dp/B0F8HX6HQZ?ref=ppx_yo2ov_dt_b_fed_asin_title&th=1

FPH Sat-1
Amp assembly

3.5MM Audio Cable

JST Connector
https://www.amazon.com/dp/B0DPF3FMJK?ref=ppx_yo2ov_dt_b_fed_asin_title

90 Degree Audio Jacks
https://www.amazon.com/dp/B09CDC1MV9?ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_13

3.5mm male both ends
https://www.amazon.com/dp/B074QHNY5Q?ref=ppx_yo2ov_dt_b_fed_asin_title

3.5mm male both ends shielded – RECOMMENTED
https://www.amazon.com/dp/B082PLZB54?ref=ppx_yo2ov_dt_b_fed_asin_title

FPH Sat-1
Amp assembly

LD2450

https://www.amazon.com/dp/B0CQ2P4RVP?ref_=ppx_hzsearch_conn_dt_b_fed_asin_title_3&th=1